Regime-Aware Machine Learning for Technical Pattern Trading

An Investigation into Event-Based Classification,
Profitability Evaluation, and Generalization on SPY Daily Data

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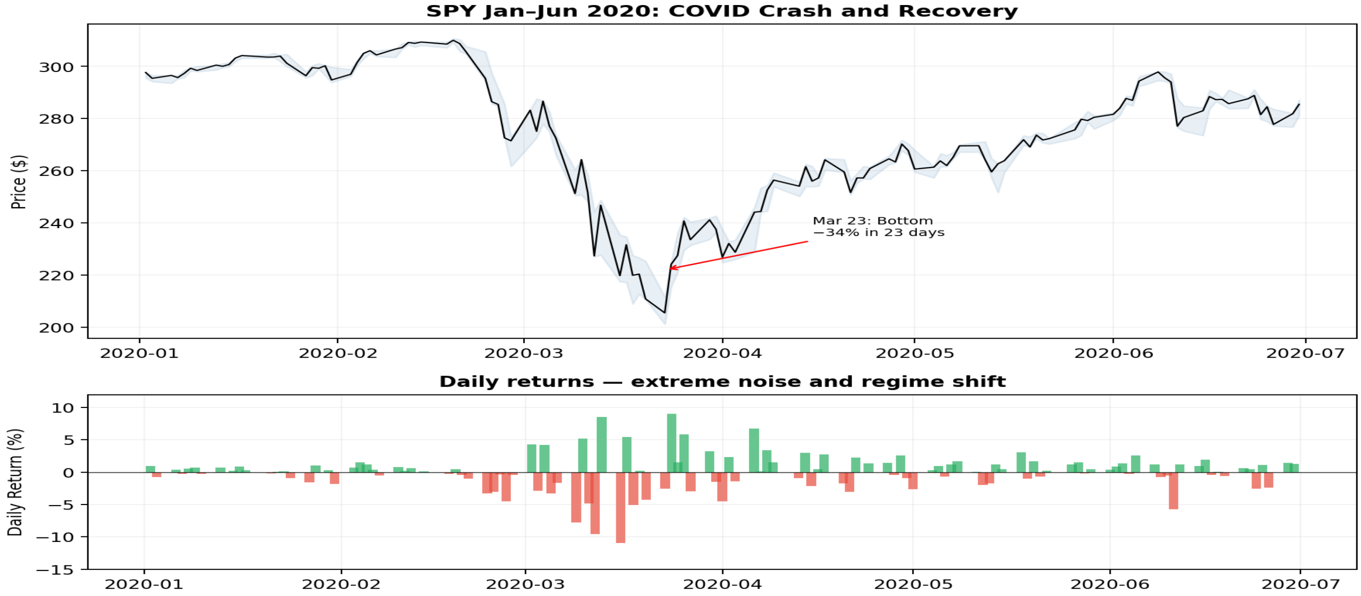
Context: Financial markets are among the hardest prediction domains in ML. Most daily price bars are noise. This project investigates whether focusing on technically meaningful events, rather than predicting every bar, can produce measurable classification and trading performance.

Research questions:

- Can pattern-filtered ML beat a random baseline on trade-outcome prediction?
- Do classification-optimal parameters also maximise trading profit?
- How stable are results across different market regimes?

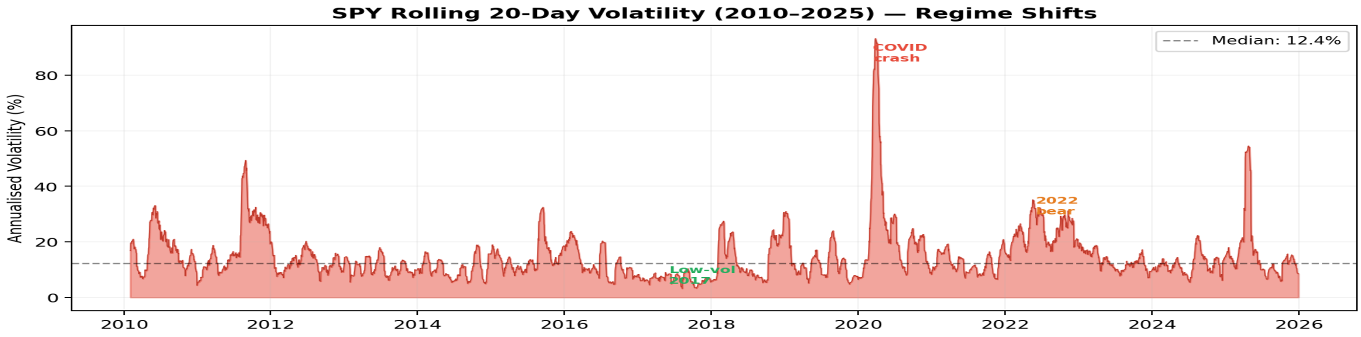
Methodology: 4 pattern detectors → 48 features → triple-barrier labels → Random Forest → walk-forward CV → profitability simulation. Evaluated on 4,023 SPY daily bars (2010-2025), producing 132 events.

The Problem: Markets Are Hostile to ML



SPY COVID crash: –34% in 23 days. Daily returns swing –12% to +9%.

The top panel shows SPY's price collapsing from $337 to $222 in just 23 trading days. The bottom panel reveals the daily return distribution during this period, a pattern no stationary model can anticipate. Most individual bars are indistinguishable from random noise.



Rolling 20-day annualised volatility across the full 2010–2025 dataset.

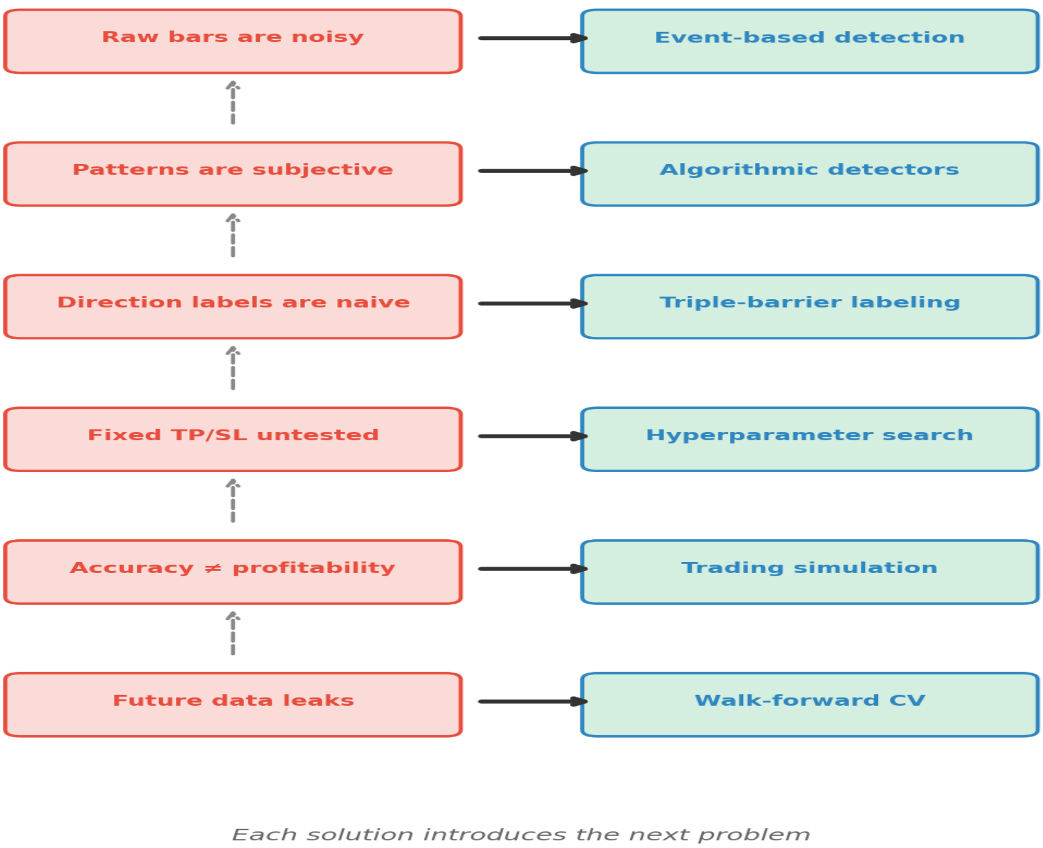
Volatility is not constant. The 2017 market had annualised vol below 5%; during COVID it spiked above 90%. A model trained during one regime faces entirely different statistical properties in the next. This non-stationarity is the fundamental enemy of financial ML.

■ ***Naive bar-by-bar models learn price trends and time artefacts rather than genuine trading edges. When the regime shifts, they fail catastrophically.***

➔ ***Our approach: detect technically meaningful events and predict only at those moments. This reduces 4,023 noisy bars to ~100 focused trading candidates.***

Research Design

Each solution creates the next problem



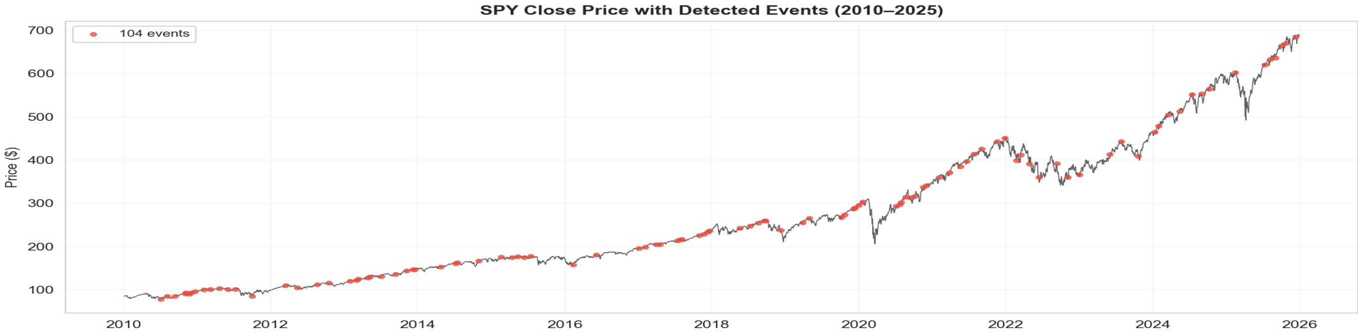
The causal chain that shaped the system design.

Read top-to-bottom: raw bars are too noisy, so we introduce event detection. Subjective patterns need algorithmic formalisation. Naive up/down labels ignore trade outcomes, so we adopt triple-barrier labeling. Fixed barrier parameters embed untested assumptions, so we optimise them. Accuracy can mislead, so we simulate trading. Standard CV leaks future data, so we use walk-forward validation. Each problem motivated the next design decision.

➔ ***The pipeline is not arbitrary, every component exists because the previous step was insufficient.***

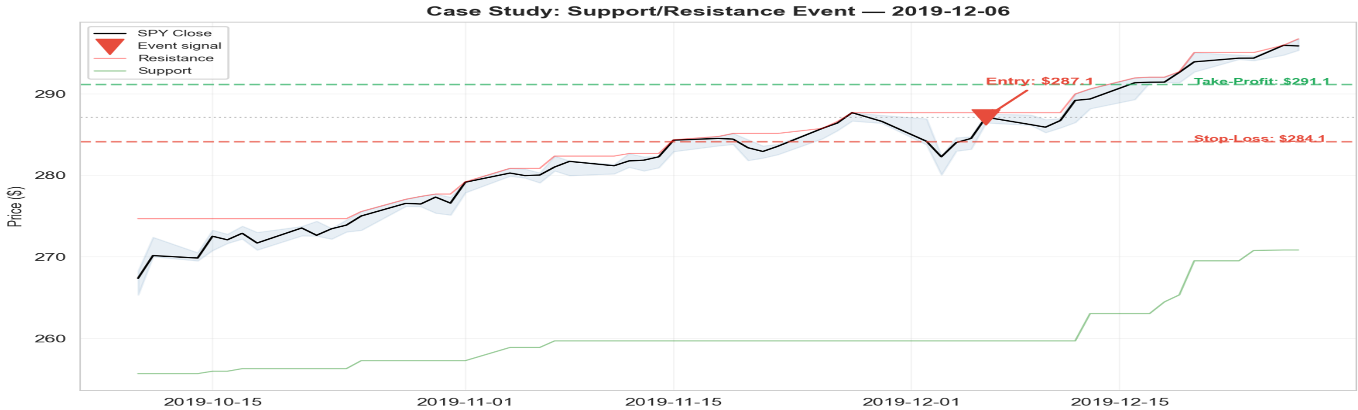
Step 1: Event Detection

What does the system actually see?



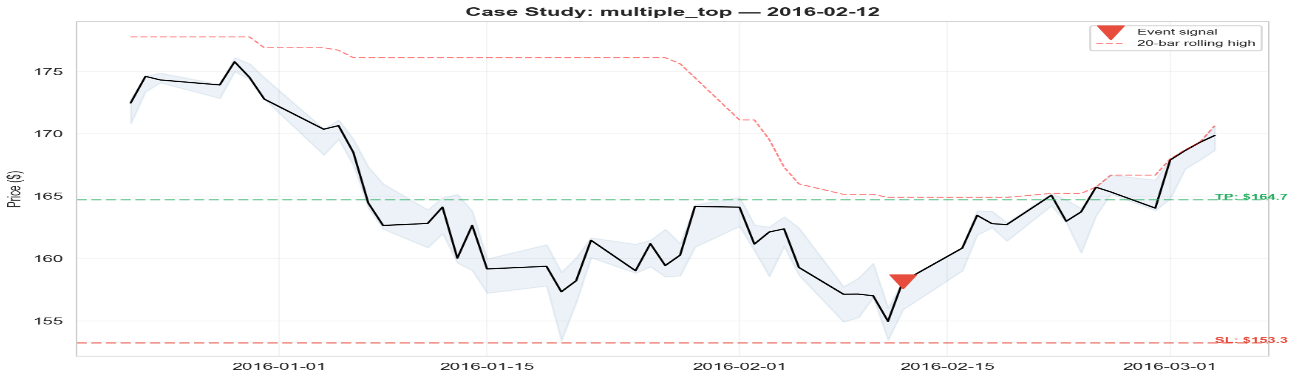
SPY close price (2010–2025) with 104 detected events overlaid in red.

The red dots are sparse and intentional. Four independent detectors (support/resistance, channels, triangles, multiple tops/bottoms) collectively flag only 3.3% of all bars. The remaining 97% is discarded as noise. This aggressive filtering is the foundation of the event-based approach.



Case study: a real support/resistance event detected on SPY.

The red triangle marks where the S/R detector fired. The green dashed line is the take-profit barrier (entry + 2.0×ATR) and the red dashed line is the stop-loss (entry – 1.5×ATR). The horizontal green line shows the support level the detector identified. This event illustrates how the system translates a technical pattern into a concrete trade setup with defined risk.



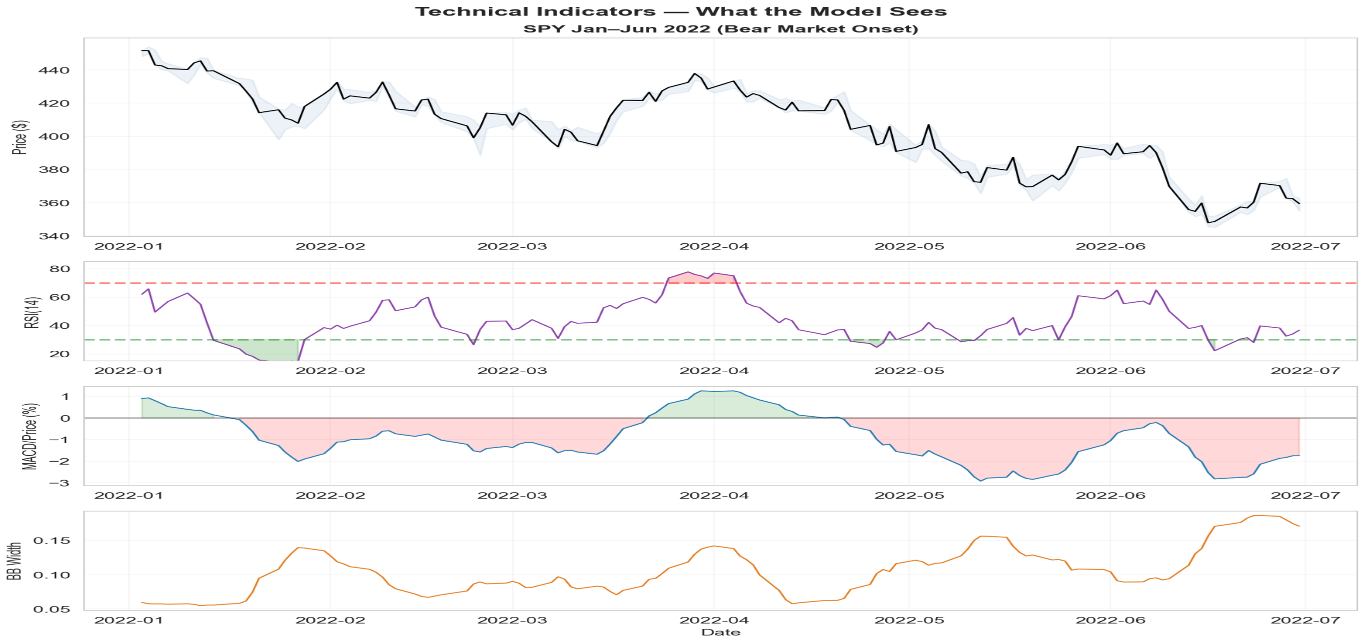
Case study: a multiple-top reversal pattern with barrier overlay.

Here the rolling 20-bar high formed a ceiling while the 5-bar close slope turned negative, a classic exhaustion reversal. The detector captures the moment where buyers can no longer push price higher, suggesting a directional opportunity.

→ 132 events from 4 detectors. Each uses ATR-based thresholds for scale independence and 10-bar cooldown to prevent duplicate signals. Triangles/channels excluded per supervisor feedback.

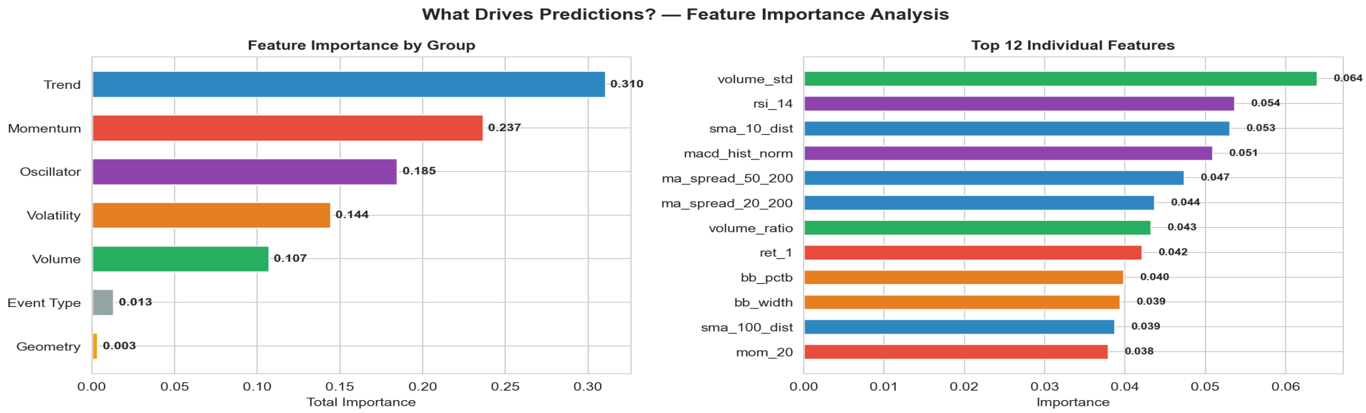
Step 2: What The Model Sees

48 features: indicators, geometry, event type



Four-panel indicator view of SPY during the 2022 bear market onset.

The top panel shows price declining from January through June 2022. The RSI panel highlights oversold zones (below 30, shaded green) where mean-reversion signals fire. The normalised MACD captures the momentum shift from positive to deeply negative. Bollinger Band width widens as volatility spikes. These are the signals the model receives at each event bar, not raw prices, but derived, bounded indicators.



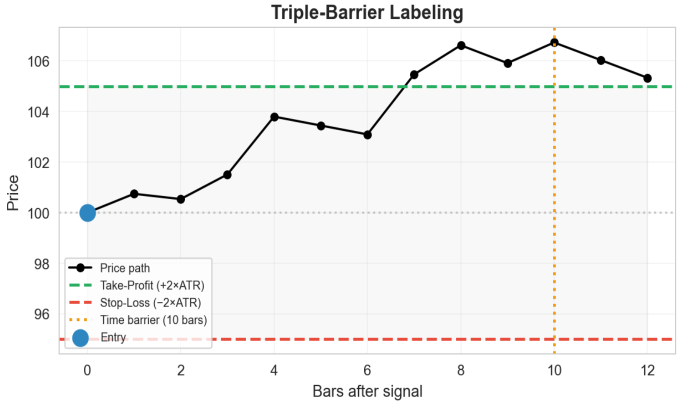
Feature importance: grouped (left) and individual top 12 (right).

Trend features (MA distances) and momentum features (returns, rate-of-change) together account for over 60% of total importance. Pattern geometry features, slopes, touches, containment, contribute less than expected. The model relies more on the *market context* around a pattern than on the pattern's geometric shape. This suggests that technical patterns may act as useful event filters, while the predictive signal comes from the surrounding market dynamics.

■ ***Removed features: raw ATR (scales with price level), absolute SMAs (act as time proxies), cumulative OBV (trends monotonically), raw MACD (price-dependent). Including any of these would allow the model to identify which time period a sample comes from, inflating test metrics.***

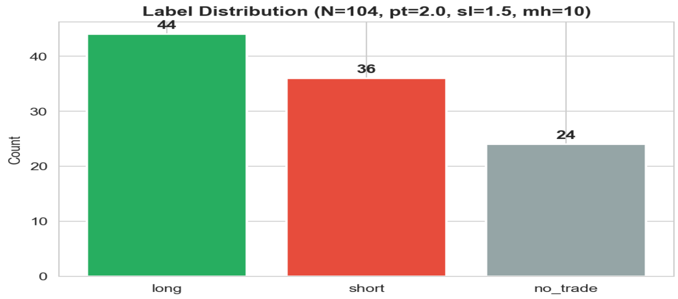
➜ ***Every feature at bar i uses only data up to bar i. Normalised values prevent temporal leakage.***

Step 3: Triple-Barrier Labels + Failed Predictions



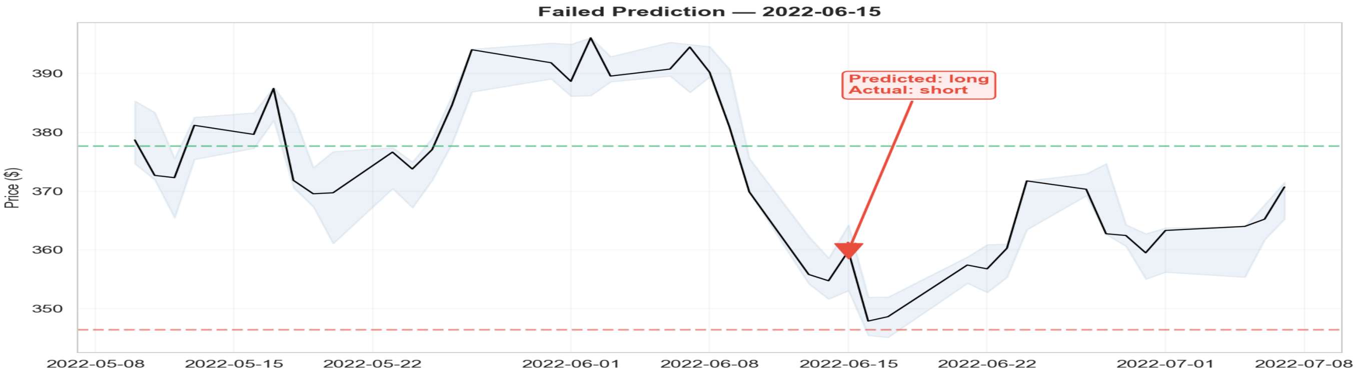
Triple-barrier labeling method (Lopez de Prado, 2018).

After each event signal, three barriers compete: the take-profit (entry + pt_mult×ATR), the stop-loss (entry – sl_mult×ATR), and the time limit (max_holding bars). Whichever is hit first determines the label. This is fundamentally different from naive up/down labeling, it encodes realistic trade outcomes with built-in risk management.



Label distribution with the best-F1 configuration (pt=2.0, sl=1.5, mh=10).

The classes are roughly balanced, though "long" slightly dominates, reflecting SPY's historical upward bias. Changing the barrier parameters shifts this distribution substantially: wider stops produce more directional labels, tighter stops produce more no_trade.



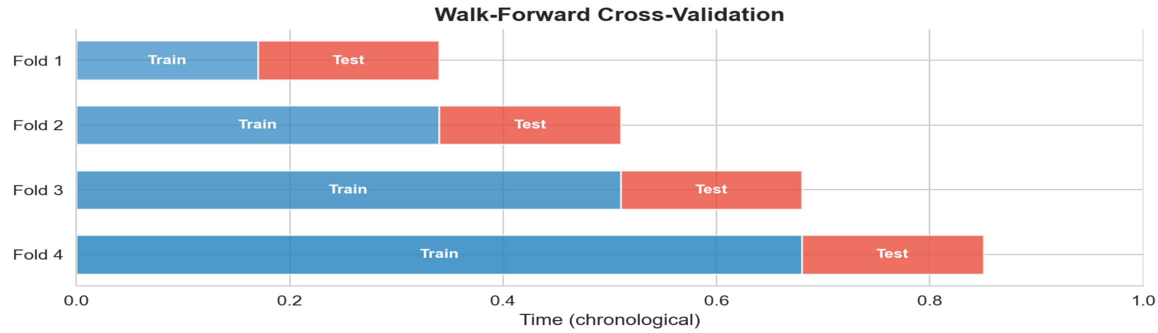
A real failed prediction from the test set.

Not every signal works. Here the model predicted incorrectly and the trade hit the stop-loss. Showing failures is essential for scientific honesty. These misclassifications are precisely why we need profitability analysis, classification metrics alone cannot tell us whether the winning trades compensate for losses like this one.

➔ ***pt_mult, sl_mult, max_holding are hyperparameters, not constants. Changing them redefines the task.***

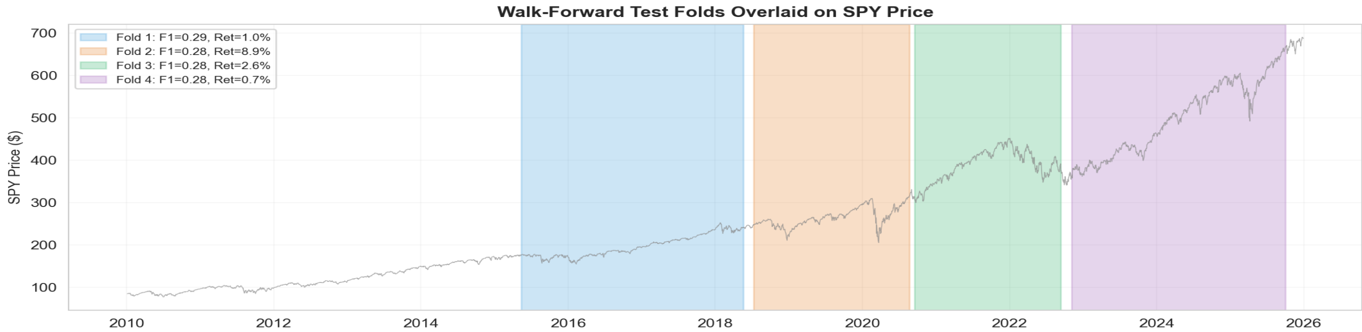
Step 4: Honest Validation

Walk-forward CV reveals what k-fold hides



Walk-forward CV: training always precedes testing in time.

In standard k-fold CV, data from 2023 can appear in the training set while data from 2018 is in the test set, the model literally sees the future. Walk-forward CV prevents this by using an expanding window: fold k trains on everything before fold k and tests on the next unseen period. This simulates how the model would perform if deployed and periodically retrained.



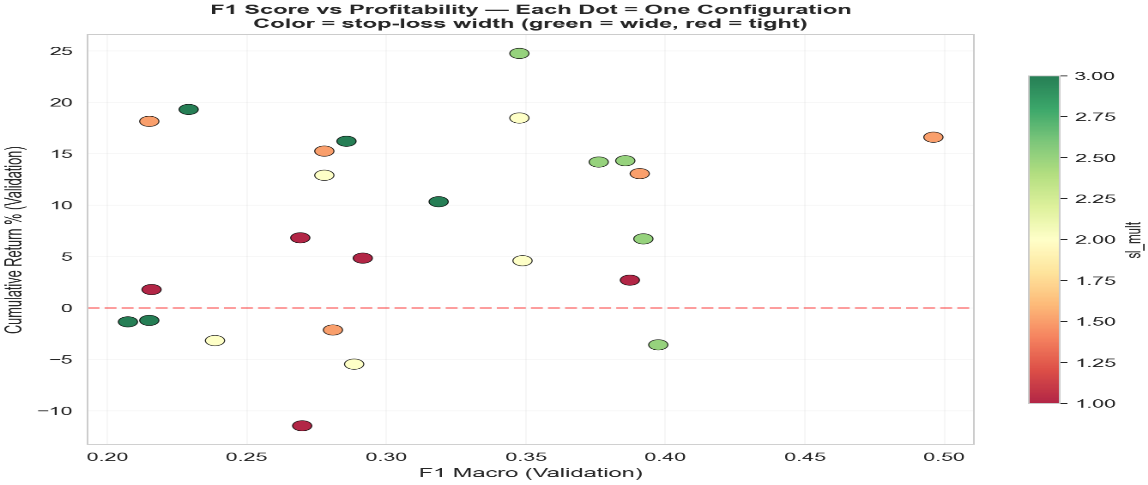
Test folds overlaid on SPY price, showing which market regime each fold covers.

The coloured bands show which price history each test fold spans. Each fold covers a different market environment: a correction, a recovery, a bear market. Performance naturally varies across these regimes, this variation is real, not an artefact. K-fold would hide it by blending events from all periods together.

Method	Temporal?	F1 variability	Purpose
Walk-forward (4 folds)	Yes	0.282 ± 0.008	Honest deployment simulation
K-fold (5 folds)	No	Lower variance	Sampling noise reduction (diagnostic only)

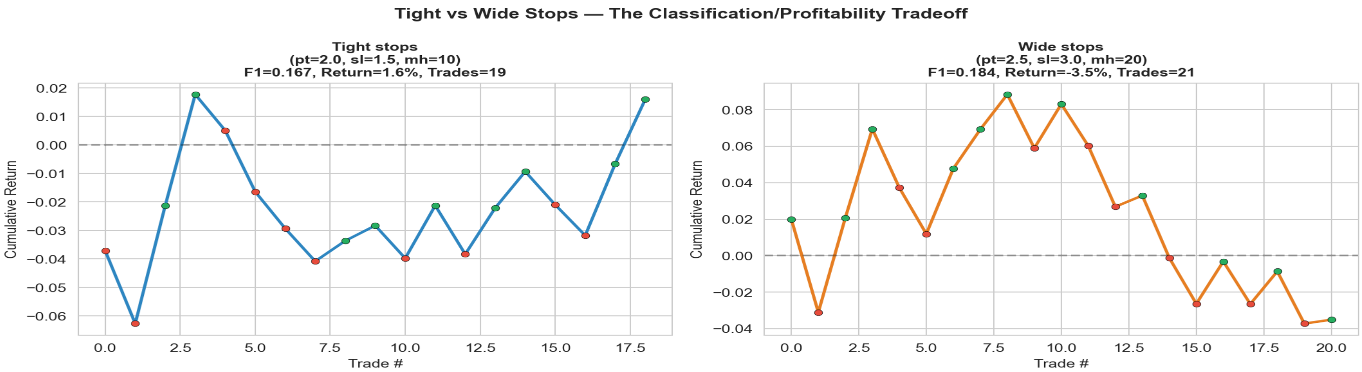
➔ ***Walk-forward is harder to pass but honest. It is the gold standard for financial ML evaluation.***

Central Finding: F1 ≠ Profitability



Each dot = one (pt, sl) configuration. Colour = stop-loss width (green = wide, red = tight).

This scatter plot is the most direct visualisation of the central finding. Points on the right have high F1 but are not necessarily at the top (high return). Green dots (wide stops, sl=2.5–3.0) cluster at higher returns despite mid-range F1. Red dots (tight stops) achieve the highest F1 scores but produce lower cumulative returns. The two objectives pull in different directions.



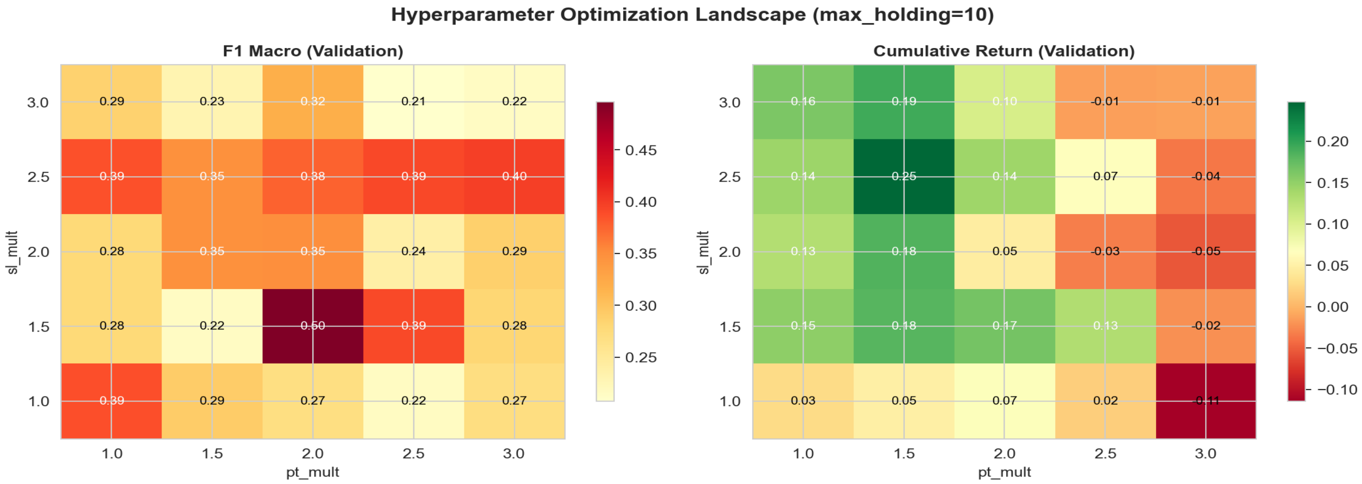
Side-by-side equity curves: tight stops (left) vs wide stops (right).

With tight stops, the model makes more trades and classifies labels more accurately, but individual gains are small because normal volatility frequently triggers the stop-loss even when the directional prediction is correct. With wide stops, the model makes fewer trades, misclassifies more events, but winning trades have room to develop into larger gains. The asymmetric reward structure favours wide stops for profitability.

→ Best F1=0.569 at pt=2.0/sl=1.5. Best return=25.9% at pt=2.5/sl=3.0. Classification and profitability optimise at different parameter settings.

Hyperparameter Landscape

100 configurations, divergent optima



Optimisation landscape: F1 (left) and cumulative return (right) across 25 (pt, sl) combinations.

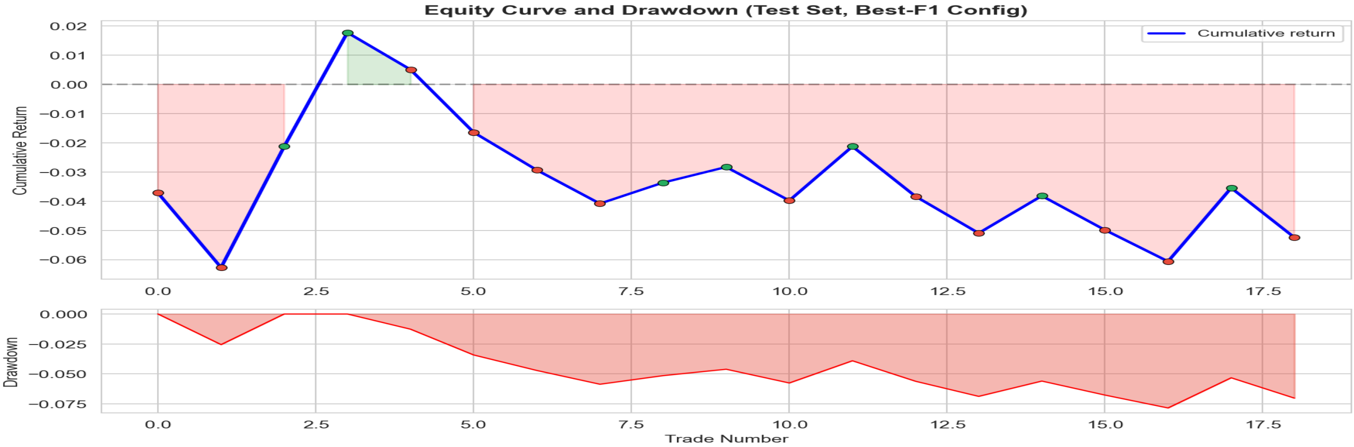
The left heatmap shows F1 performance; the right shows cumulative return. The brightest cells in each map occupy different positions. High F1 concentrates at moderate pt with tight sl (lower rows). High return requires wider sl (upper rows), where the model misclassifies more events but the profitable trades generate larger gains per trade.

Config	pt / sl / mh	F1	Return	Win Rate	Trades
Default	2.0 / 2.0 / 10	0.160	3.7%	50%	18
Best F1	2.0 / 1.5 / 10	0.569	8.5%	55%	18
Best Profit	2.5 / 3.0 / 20	0.392	25.9%	52%	15

The default configuration achieves F1 of just 0.16, barely above random. Optimisation lifts this to 0.569, a 3.6× improvement. But the most profitable configuration (F1=0.392) uses entirely different barrier parameters. These parameters do not merely tune the model, they *redefine the classification task itself*. A "long" label with tight stops is a fundamentally different outcome than a "long" with wide stops.

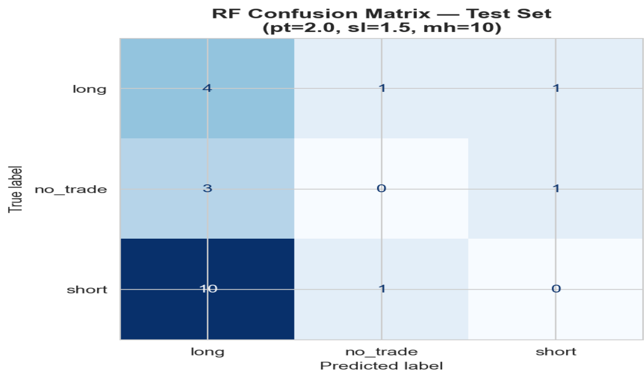
Profitability Analysis

Equity curves, drawdowns, trade fragility



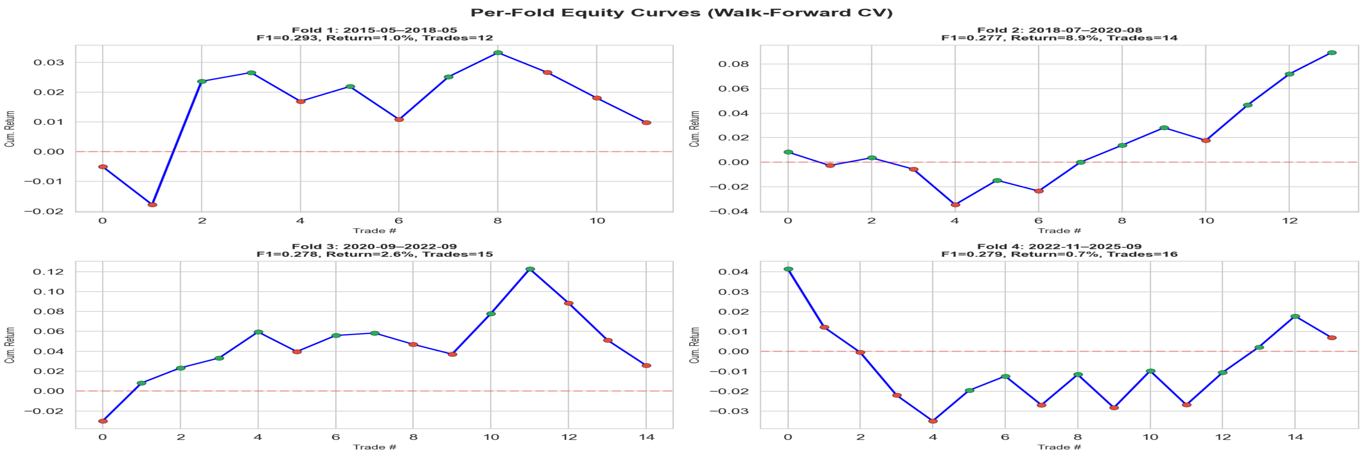
Equity curve (top) and drawdown (bottom) on the held-out test set.

Green dots mark winning trades; red dots mark losers. The strategy accumulates gains through a handful of larger winners, but the drawdown panel shows periods where those gains are partially given back. A few dominant trades drive the overall return, removing the top 2–3 winners would dramatically change the picture. This concentration risk is typical of small-sample trading strategies.



RF confusion matrix on the test set (best-F1 parameters).

The model over-predicts "long", consistent with SPY's historical upward bias over 2010–2025. Short predictions are both rare and frequently incorrect. Learning to predict shorts is harder because bearish patterns on a long-term bullish asset are intrinsically less common and more noisy. The no_trade class is also challenging, as the model tends to take a directional position rather than abstain.



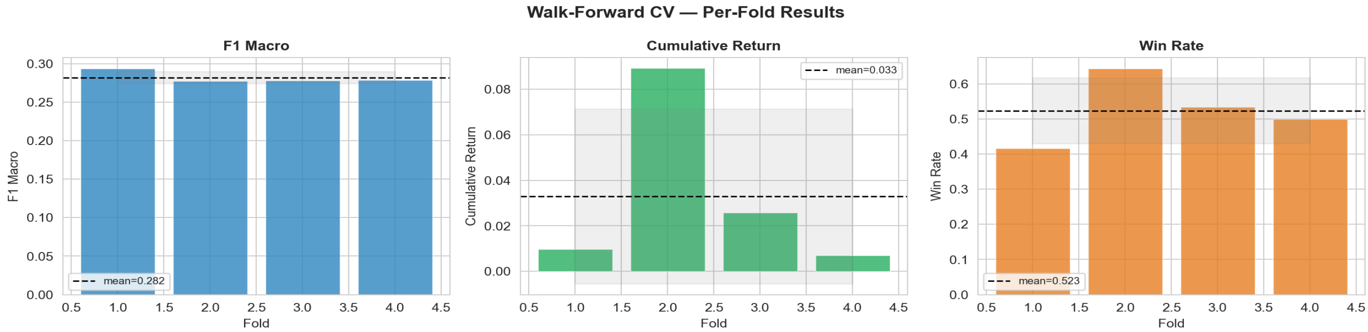
Per-fold equity curves from walk-forward CV.

Each subplot shows the equity curve for one walk-forward fold. The variation is substantial: some folds produce positive returns, others barely break even. With only ~17 test events per fold, a single large trade can swing the entire fold outcome. This instability is not a model flaw, it is the unavoidable consequence of working with a small event dataset.

→ ***Profitability is fragile and regime-dependent. The edge is real but thin.***

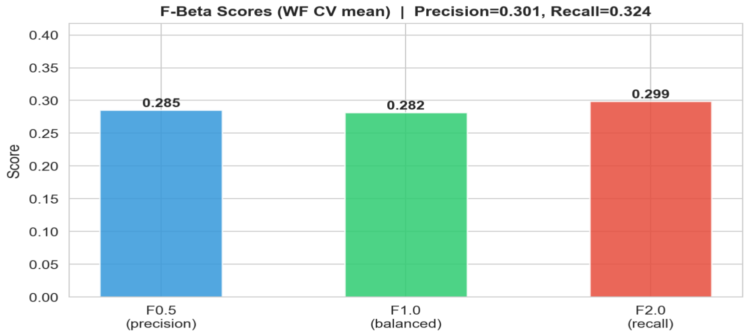
Generalization and F-Beta

How uncertain are these results?



Per-fold metrics from walk-forward CV (mean shown as dashed line, ±1 std as grey band).

F1 is the most stable metric (std=0.008), but cumulative return swings from near-zero to 7% depending on the fold. Sharpe ratio varies between 0 and 0.3. Win rate hovers around 50% with substantial uncertainty (±9%). These wide confidence bands mean that the reported single-split results (F1=0.569, return=25.9%) represent favourable realisations from a broad distribution, not stable point estimates.



F-beta scores averaged over walk-forward folds.

F0.5 (precision-heavy) = 0.285, F1 (balanced) = 0.282, F2 (recall-heavy) = 0.299. Precision (0.30) and recall (0.32) are nearly balanced. The model is not strongly biased toward either avoiding false alarms or capturing all opportunities. In trading terms, a precision-focused strategy (F0.5) would take fewer but higher-confidence trades; a recall-focused strategy (F2) would trade more aggressively, accepting more losses in exchange for not missing profitable setups.

Error Type	Trading Meaning	Cost
False Positive	Enter trade → lose money	Direct financial loss
False Negative	Skip trade → miss profit	Opportunity cost (no loss)

➜ ***Walk-forward variance is the most important diagnostic in this project. It reveals the uncertainty that single-split evaluation hides.***

What We Learned

1. Event filtering works. Reducing 4,023 bars to 132 events concentrates the model on moments with structure. RF achieves 3.6× F1 improvement over the stratified random baseline (0.569 vs 0.160). The 97% of bars discarded as noise would have diluted this signal.

2. Classification ≠ profitability. The best F1 configuration uses tight stops (sl=1.5×ATR, 10-bar holding), producing accurate but small trades. The best profit configuration uses wide stops (sl=3.0×ATR, 20-bar holding), allowing winners to develop despite lower classification accuracy. This divergence is confirmed across all walk-forward folds and the full optimisation heatmap.

3. Honest validation is essential. Walk-forward CV reveals F1 variance of ±0.008 and return variance of ±3.8%, showing substantial regime dependence. K-fold CV hides this instability by mixing time periods. Financial ML research that relies solely on k-fold risks dangerous overconfidence.

4. The signal is real but fragile. Technical patterns contain measurable predictive information, but the edge is thin. With ~17 test events per walk-forward fold, a single trade can swing the entire fold outcome. The framework is validated; the dataset is too small for production conclusions.

Limitations	Future Work
Small dataset (~100 events)	Multi-asset testing (ETFs, stocks)
Single asset (SPY only)	Transaction cost modeling
No transaction costs	Purged/embargo cross-validation
Optimisation overfitting risk	Regime-aware dynamic TP/SL

Thank you. Questions?